STATS 7022 - Data Science PG
Assignment 2

Trimester 1, 2025

Question 1: Recreating an Analysis

Data scientists are often asked the following question by their colleagues (especially domain experts):

> *"I found this analysis performed by another data scientist. Can you check it for me?"*

Generally, the best way to "check" the analysis performed by someone else is to repeat their entire analysis. Doing so gives you the best possible understanding of what they did and why they did it. In this question, we will recreate a short data analysis report prepared by another data scientist.

Question

You have been provided with a data analysis report produced by another data scientist. The data scientist who prepared the report has forgotten to include their code. You task is to recreate the report, making sure to include all code and all relevant output.

A copy of the `bird_bath` data used in the analysis is available on MyUni, or can be downloaded here: `bird_bath.rds`

A copy of the original data analysis report prepared by the other data scientist is available on MyUni, or can be downloaded here: `DS_A2_Q1.html`

For further information, see the assessment rubric on MyUni.

[Question total: 12 marks]

Submission

A single html file containing your recreation of the original data analysis report, including all code and relevant output.

Checklist

Please ensure that:

- Your submission includes all `R` output and plots to support your answers where necessary.
- Your submission includes all `R` code.
- Your submission includes a caption for every plot and table.

- Your submission is a single html file - correctly oriented, clear, and legible.
- You have submitted your solution by online submission to the correct portal on MyUni. Submissions will not be marked and will receive a grade of 0 if they are not made through MyUni or if they are not made to the correct portal.

The course's regular late policy applies to this assignment. If you need to request an extension, or request a variation to the assessment on medical or compassionate grounds, please contact the course coordinator.

Question 2: Mathematics of Bootstrapping

An important part of being a data scientist is understanding the mathematics that underpins the methods we use. Recognising when and how a method can go wrong forms part of this understanding. In this question, we will look at the mathematics behind bootstrapping.

Question

Bootstrapping is a resampling method that is commonly used to estimate quantities that are difficult to find analytically. Bootstrapping relies on our original sample being representative of the population from which it was drawn. What happens if our original sample includes points that are not representative of the population?

Consider a random sample $X_1, \ldots, X_n$ such that

$$E[X_i] = \begin{cases} \theta - \phi,\ \phi > 0, & i = 1, \ldots, \ell \\ \theta, & i = \ell + 1, \ldots, n. \end{cases}$$

Let S_j^* be the j^{th} bootstrap sample and $\bar{X}_j$ be the sample mean of the j^{th} bootstrap sample.

(a) Calculate the probability that S_j^* contains none of $X_1, \ldots, X_\ell$.

(b) Calculate the expected value of $\bar{X}_j$ if k of the observations in S_j^* are drawn (with replacement) from $X_1, \ldots, X_\ell$.

(c) Hence, find the value of k for which $\bar{X}_j$ is an unbiased estimator for θ.

(d) Calculate the expected bias of $\bar{X}_j$ as an estimator for θ.

[Question total: 10 marks]

Submission

A single pdf file containing your solutions. You may handwrite your answers and scan them, or typeset your answers.

Checklist

Please ensure that:

- Your submission is a single pdf file - correctly oriented, clear, and legible.
- You have shown all of your working, including probability notation where necessary.

- You have submitted your solution by online submission to the correct portal on MyUni. Submissions will not be marked and will receive a grade of 0 if they are not made through MyUni or if they are not made to the correct portal.

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Question 3: LDA Function

One of the best ways to understand an algorithm is to implement it yourself. In this question, we will write a function to perform Linear Discriminant Analysis (LDA) with two predictor variables.

Question

In `R`, write a function called `get_lda2()`. Your function should take five vector inputs:

- `x1`: a numeric vector of the first predictor.
- `x2`: a numeric vector of the second predictor.
- `y`: a character vector of classes (the response).
- `new1`: a numeric vector of `new_data` for the first predictor.
- `new2`: a numeric vector of `new_data` for the second predictor.

Your function should return a vector of predicted classes for `new1` and `new2`.

Here is an example to illustrate its use:

```
x1 <- c(5,10,11,4,6,9,3,7,8)
x2 <- 8:0
y <- rep(LETTERS[19:17], each = 3)
new1 <- c(6,5)
new2 <- c(2,8)
get_lda2(x1 = x1, x2 = x2, y = y, new1 = new1, new2 = new2)
```

```
## [1] "Q" "S"
```

Here is a template you can use:

```
# Function to perform LDA
#
# INPUT
# Five vectors:
#   x1: first predictor
#   x2: second predictor
#   y: classes
#   new1: new first predictor
#   new2: new second predictor
#
# OUTPUT
# A vector of predicted classes for new1 and new2.

get_lda2 <- function(x1, x2, y, new1, new2) {
```

```
    # YOUR CODE HERE
}
```

Your function should also ensure that the correct type of input is passed to the function and that edge cases are handled appropriately. For further information, see the assessment rubric on MyUni.

[Question total: 9 marks]

Submission

A single `R` file containing your function. Your code will be automatically unit-tested, so any changes to names etc. may result in a loss of marks. Your function must not use LDA functions from other packages - the code must be your own work.

Checklist

Please ensure that:

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Question 4: Technical Communication

There are four key skills in a data scientist's arsenal: analysis, mathematics, coding, and communication. In this question, we'll practice your communication skills.

Whenever you are communicating, it's important to consider your audience - different audiences will need to know different information, and be able to understand different levels of technical detail.

Question

Suppose you are a professional data scientist who has been asked to write a brief report explaining Principal Component Regression (PCR) for a junior colleague. You may assume that the junior colleague has about the same level of knowledge about data science as a student in STATS 7022 Data Science PG (except that they are unfamiliar with principal component regression!). Your report should address the following points:

- what sort of model PCR is, and what types of data it can be used with;
- how PCR works; and
- the advantages and disadvantages of PCR.

Marks will be allocated for the quality and correctness of your writing, as well as how suitable your writing is for the intended audience. Your report must use Harvard-style referencing as detailed here:
https://www.adelaide.edu.au/library/referencing-support

For further information, see the assessment rubric on MyUni.

[Question total: 13 marks]

Submission

A single doc, docx, or pdf file containing your report. Your report should be 1-2 A4 pages (including referencing), with 10-12pt font, single line spacing, and standard page margins.

Checklist

Please ensure that:

- Your submission is a single doc, docx, or pdf file - correctly oriented, clear, and legible.
- Your submission is not longer than 2 pages (including tables, figures, and references).
- Your submission is correctly referenced. For more information, see the link above or visit the Writing Centre.

- You have submitted your solution by online submission to the correct portal on MyUni. Submissions will not be marked and will receive a grade of 0 if they are not made through MyUni or if they are not made to the correct portal.

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